

AARUSH SINGH

Graduate Software Engineer | Full-Stack · Cloud · IT Support | Melbourne, Australia

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SUMMARY

Software Engineering graduate from Swinburne (First Class Honours). I've built production systems that real people use (a restaurant ordering platform, fintech payment flows, autonomous robotics on hardware) and I'm comfortable diagnosing and fixing production incidents under pressure. I know React, FastAPI, PostgreSQL, and AWS, but what matters to me is shipping software that works reliably and solving real problems for actual users. Currently supporting Uber operations at Melbourne Airport.

EDUCATION

Swinburne University of Technology

Feb 2022 – Dec 2025

Bachelor of Engineering (Honours) – Software Engineering | GPA: 3.2 / 4.0 | Melbourne, Australia

Coursework: Distributed Systems, Cloud Computing (AWS), Database Management, Computer Networks, Software Architecture, Linux Systems

Awards: International Excellence Scholarship (75%), Industry Experience Certificate, Swinburne Emerging Leader, Sustainability Certificate

TECHNICAL SKILLS

Languages: Python, JavaScript (ES6+), TypeScript, SQL, Java, C#, C++, HTML5, CSS3

Web & Frameworks: React.js, React Native, FastAPI, Node.js/Express, Flask, ROS2, Tailwind CSS

Databases: PostgreSQL (Supabase), MySQL, MongoDB, Firebase – schema design, querying, data modelling

Cloud & DevOps: AWS (CloudWatch, EC2, S3), Microsoft Azure, Docker, GitHub Actions (CI/CD), Vercel, Render, Git

IT Support & Tools: Hardware troubleshooting, AV systems, Jira, ServiceNow, incident triage, escalation documentation

AI / ML: CNNs, LSTMs, sensor fusion, OCR (Tesseract, EasyOCR), TensorFlow, OpenCV, Claude API, GitHub Copilot

APIs & Integration: RESTful APIs, Webhooks, JWT, Stripe, Postman, HTTP/HTTPS debugging, TLS/SSL, PKI

Testing & Quality: PyTest, Jest, integration testing, unit testing, code review

WORK EXPERIENCE

Airport Operations Specialist

Jun 2022 – Present

Uber | Melbourne Airport, VIC

- Supporting hundreds of riders and drivers every shift by troubleshooting app issues, including payment problems, login failures, and driver-rider matching, helping people get moving efficiently. Every day, I gain firsthand insight into what works and what doesn't in a high-pressure, real-time environment.
- Monitoring real-time operations during peak hours and managing the Perfect Match system to keep rider-to-bay assignments running smoothly. Helped pilot and support the rollout of Uber Kiosks, Uber Child Seat, and UberMax, with all initiatives tested first at Melbourne Airport before deployment across the ANZ network.
- Providing field feedback to the product team by identifying issues firsthand and translating observations into structured recommendations that contribute to product improvements deployed across the region. Developed a strong understanding of how frontline operations influence decisions affecting the broader network.

Autonomous Robotics Capstone Project

Feb 2025 – Nov 2025

Swinburne University of Technology, Faculty of Engineering | Melbourne, VIC

- Developed autonomous navigation algorithms in Python and C++ using ROS2 Nav2, achieving 95%+ accuracy on real robot hardware through iterative real-world testing across a warehouse simulation environment.
- Designed a sensor fusion pipeline integrating LiDAR, wheel encoders, and camera data with real-time filtering and anomaly detection, maintaining stable localisation under dynamic conditions; built a dual-engine OCR system (Tesseract and EasyOCR) with confidence scoring and fuzzy matching for reliable warehouse signage recognition.
- Diagnosed cross-subsystem failures spanning SLAM, navigation, and sensor layers; delivered structured technical progress reports and corrective action summaries to a senior academic client across a 10-month engagement.

Software Engineer Intern

Jul 2024 – Feb 2025

YEMEC Company Pty Ltd | AI-powered fintech platform – split payments and loyalty automation | Melbourne, VIC

- Diagnosed and resolved live production incidents involving Stripe API failures and webhook synchronisation errors by analysing AWS CloudWatch logs and tracing HTTP request flows; tracked all defects in Jira with reproduction steps, severity assessments, and escalation notes for senior engineers.
- Built and maintained RESTful API endpoints and webhook integrations in Node.js and Express with a PostgreSQL backend; delivered a vendor dashboard with real-time order updates and Stripe payment processing, and configured Docker, GitHub Actions CI/CD, and CloudWatch monitoring across the production environment.
- Shipped a full React Native mobile application end-to-end covering vendor authentication, flyer customisation, and loyalty reward flows; served as the primary technical contact for vendor stakeholders, translating incident findings into plain language and maintaining progress throughout resolution.

PROJECTS

Full-Stack Restaurant Ordering Platform | React, FastAPI, PostgreSQL, Stripe, Cloudinary, Vercel

mohrstreetcafe.com.au

- Designed and shipped a complete production ordering system for a live cafe, featuring QR table ordering, real-time order tracking, an admin dashboard, 30+ delivery zones, customer loyalty points, thermal receipt printing, and Stripe payment integration.
- Architected a PostgreSQL schema on Supabase to manage orders, menus, delivery zones, and loyalty data; deployed across Vercel and Render with Cloudinary for media storage, maintaining zero-downtime releases throughout.
- Handled all ongoing production responsibilities independently – incident monitoring, bug triage, schema updates, and feature releases – for a system actively serving real customers.

Full-Stack E-Commerce Web Application | Flask, React.js, MySQL, Tailwind CSS

[GitHub](#)

- Built a retail operations platform covering product catalogue browsing, inventory management, order tracking, and separate authentication flows for staff and customers.
- Designed a normalised MySQL schema managing 20+ product records with consistent data integrity across stock levels, order history, and user accounts.
- Connected a Flask REST backend to a React and Tailwind CSS frontend, then validated the full-stack architecture with integration tests across all core user flows before deployment.

Facial Recognition Attendance System | Python, Flask, React, OpenCV, CNN, TensorFlow

[GitHub](#)

- Built a secure attendance management system using CNN-based facial recognition with anti-spoofing through blink detection and emotion recognition via OpenCV, achieving 95%+ liveness verification accuracy on test data.
- Developed a Flask REST backend to manage user authentication, registration, and attendance records, with a React frontend handling live webcam capture and user enrolment.
- Validated the full-stack architecture with integration tests across enrolment, recognition, and attendance flows before deployment to ensure reliability in a real-world environment.

Full-Stack Personal Finance Web App | React.js, Node.js, MongoDB, Azure, Docker, GitHub Actions

[GitHub](#)

- Built a personal finance application with real-time analytics dashboards, interactive spend visualisations by category, and per-user data isolation via JWT-based authentication with protected routes.
- Containerised the application with Docker and deployed to Microsoft Azure via an automated GitHub Actions CI/CD pipeline, achieving zero-downtime releases across updates.
- Configured production monitoring with logging, health checks, and alerting to surface issues early, with a MongoDB backend handling all CRUD operations through RESTful API endpoints.

CERTIFICATIONS & WORK RIGHTS

Temporary Graduate Visa (Subclass 485) – Full work rights, valid until 19 March 2028

Working With Children Check (WWCC) – Current and valid

Certificates: Claude Code 101 (Anthropic, 2026) · Claude 101 (Anthropic, 2026) · Python 3 (Udemy, 2020) · Python (Kaggle, 2020) · SQL (Kaggle, 2020) · Introduction to Machine Learning (Kaggle, 2020)

References available upon request.